

STC Jawwwy

**Virtual Work Experience
with (STC TV) via Misk
Platform**

The logo for STC TV is displayed on a large purple circular background. The letters 'stc' are in a white, lowercase, sans-serif font, while the letters 'tv' are in an orange, lowercase, sans-serif font.

stc tv

Introduction

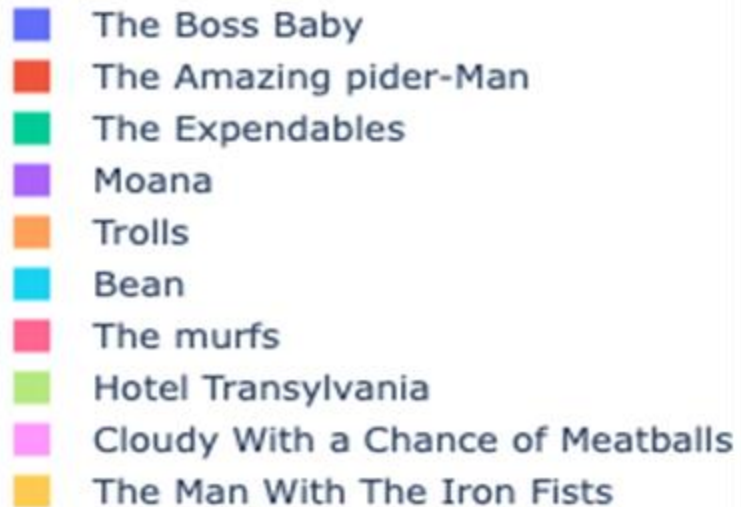
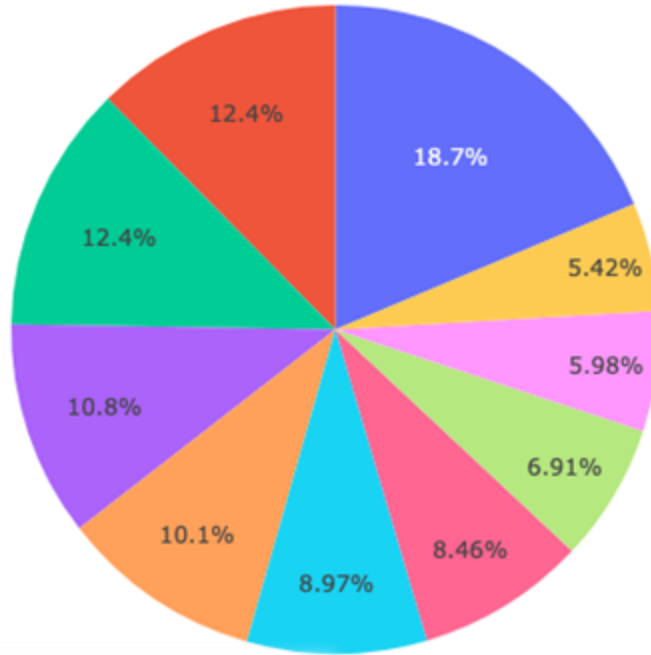
- STC is one of the leading national companies providing digital solutions and services in several fields, including information technology, communications, and entertainment services.
- As a digital company, STC has a dedicated data analysis sector. This sector includes important sections such as the advanced analysis section, which will be my field of work today.
- In this experiment, I will work with them as a data scientist and analyst, focusing on the STC TV service. My goal will be to help the company improve its performance, determine the most effective ways of working, and analyse customer interests and satisfaction to create better products and services.

Jawwy Dataset

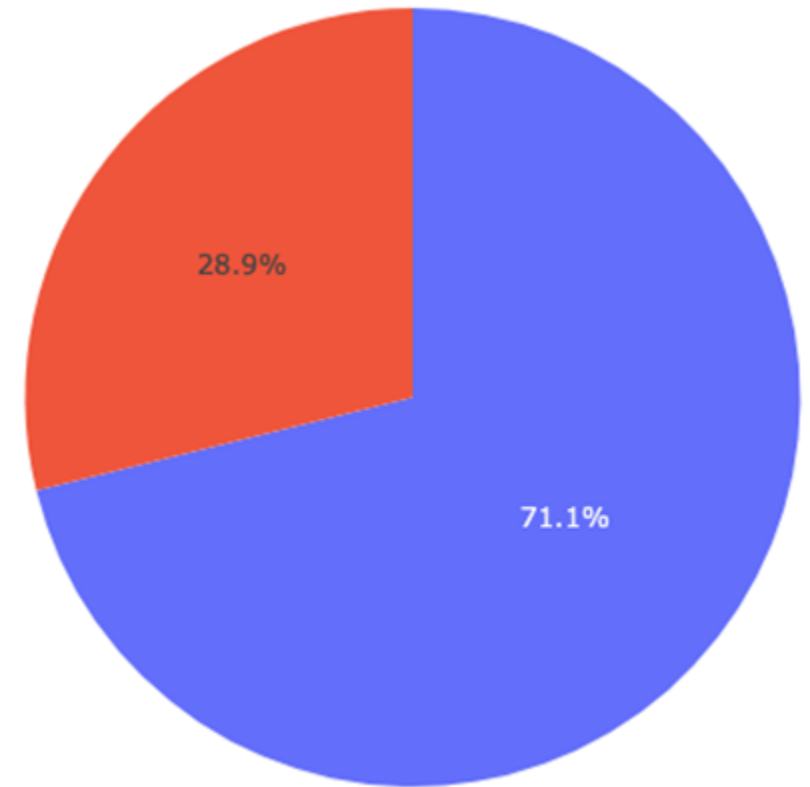
The dataset includes metadata about movies and TV shows, such as genres. It also contains details about user activities, including the duration spent watching and whether the content was viewed in high definition or standard definition. Our task is to analyse this dataset to uncover key insights and findings, and to complete the four assigned tasks.

[30]:

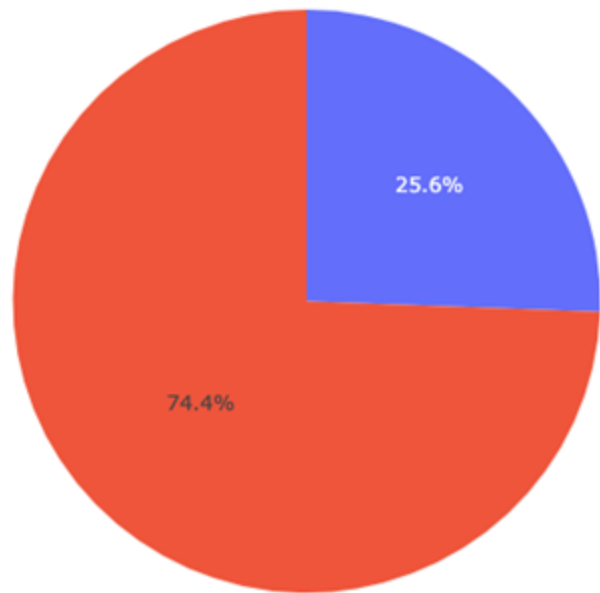
	program_name	program_class	No of Users who Watched	No of watches	Total watch time in hours
0	The Boss Baby	MOVIE	3389	24047	2961.350833
1	The Amazing spider-Man	MOVIE	1011	2877	1966.119167
2	The Expendables	MOVIE	853	2119	1961.159444
3	Moana	MOVIE	2173	8081	1706.176944
4	Trolls	MOVIE	2613	13793	1601.023056
5	Bean	MOVIE	949	3617	1423.955000
6	The mürfs	MOVIE	867	3132	1342.141111
7	Hotel Transylvania	MOVIE	491	1947	1096.533611
8	Cloudy With a Chance of Meatballs	MOVIE	683	2076	948.674722
9	The Man With The Iron Fists	MOVIE	707	2505	859.626389
10	Salt	MOVIE	563	1082	767.392778
11	Unbroken	MOVIE	625	1429	763.078333
12	ParaNorman	MOVIE	614	1746	747.065556
13	Youn Maloosh Lazma	MOVIE	1131	2278	718.109722
14	Ferdinand	MOVIE	1278	6817	714.223056
15	White Chicks	MOVIE	307	916	711.840833
16	Jurassic Park	MOVIE	504	1192	693.394444
17	The November Man	MOVIE	494	1219	679.492222
18	Total Recall	MOVIE	587	1108	661.820000
19	Robin Hood	MOVIE	588	1209	643.935000
20	Public Enemies	MOVIE	368	716	634.035000
21	Daddy Day Camp	MOVIE	263	647	625.338333



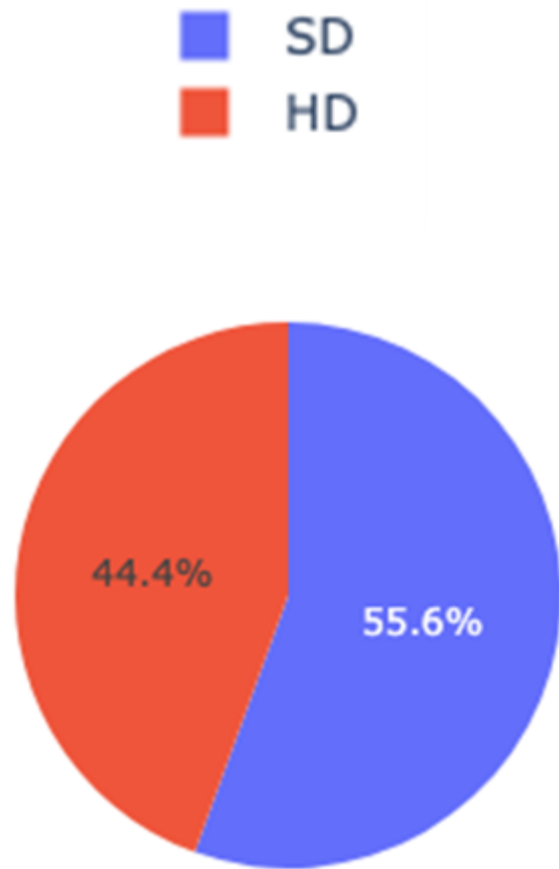
- We aim to identify the most-watched program shows by rearranging the dataset to reflect the total number of watches and the number of users who watched them over a time series. This analysis will highlight the top 10 program shows based on the total hours watched.



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- We studied the customer experience in relation to the Program class to illustrate the total duration spent, as shown in this image.

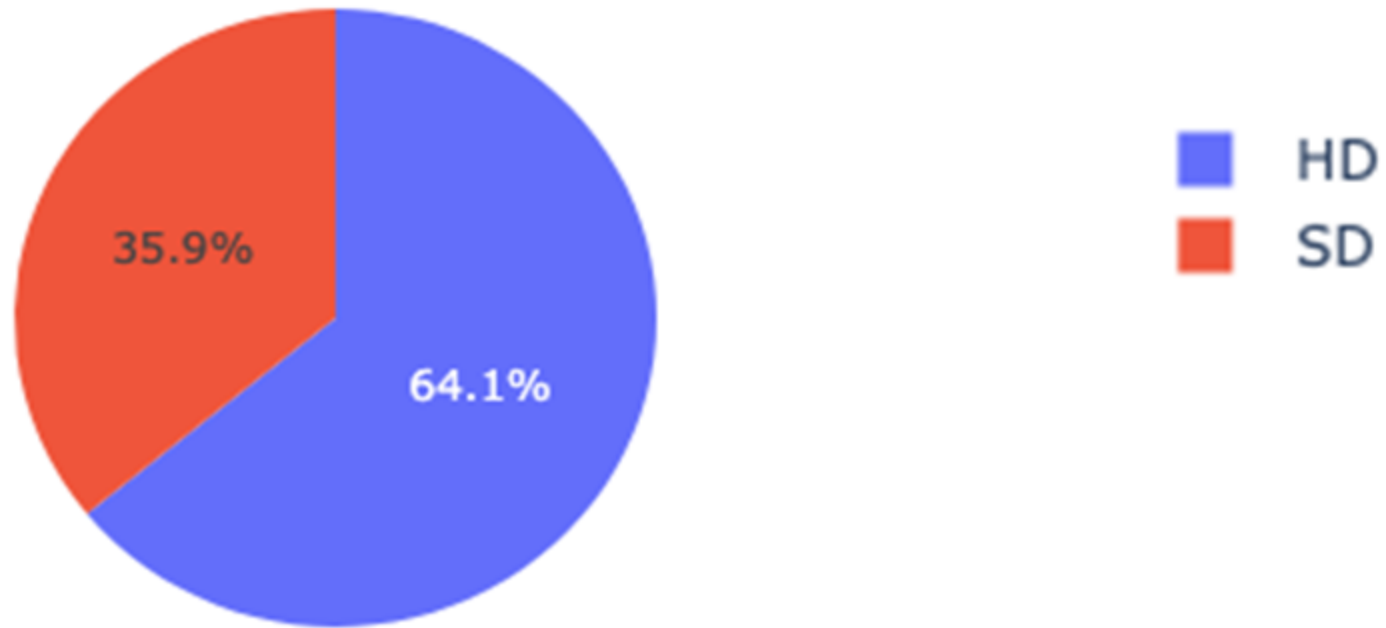


- We analysed the customer experience related to the Program class to demonstrate the total number of viewers, as depicted in this image.



- Now, We need to focus on Task One, which involves studying the HD flag in the available dataset.
- First , Series Viewers

Second : Movies Viewers



Second Task

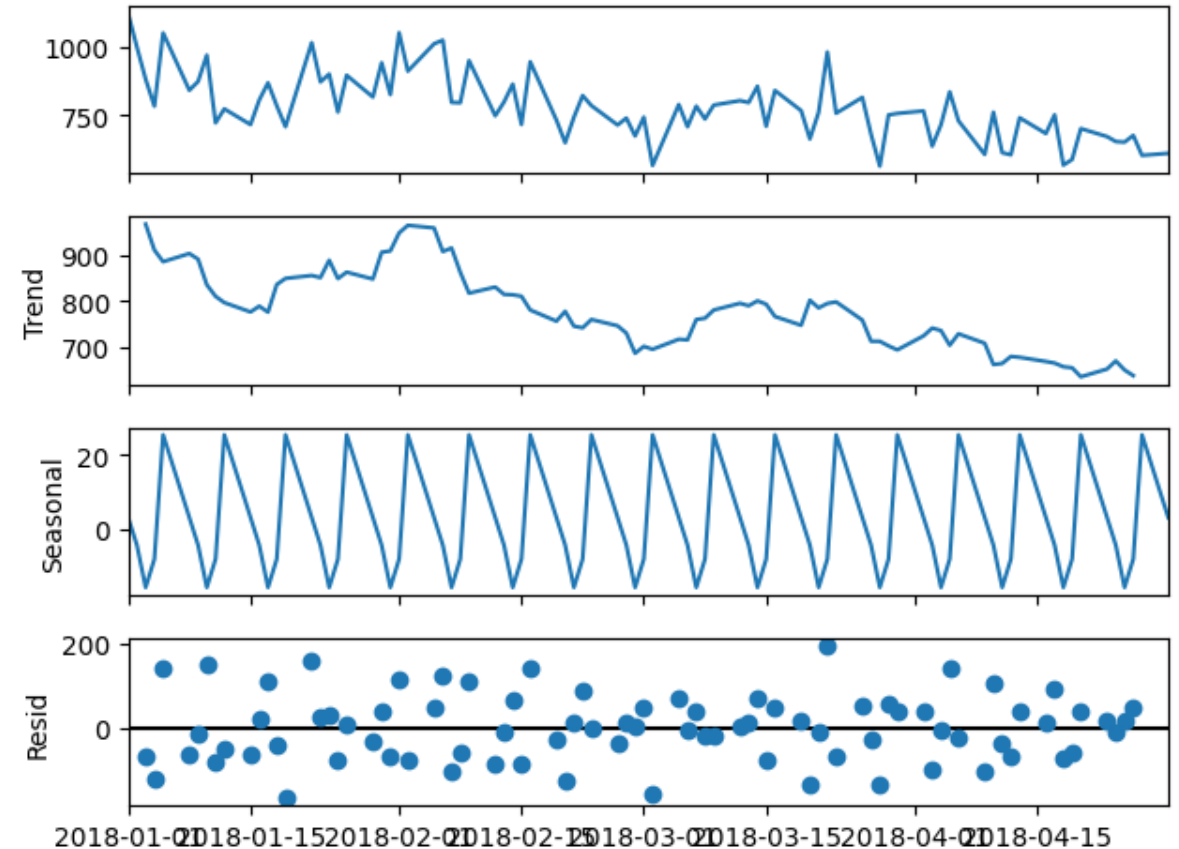
- We have a dataset that includes the total watching hours for customers per day, and we need to work on predicting the forecast for these watching hours.

Total_watch_time_in_houres	
date_	
2018-01-01	1123.551944
2018-01-02	1000.129722
2018-01-03	881.924444
2018-01-04	782.669444
2018-01-05	1051.939444
2018-01-08	841.178333
2018-01-09	873.495000
2018-01-10	970.475000
2018-01-11	721.381944
2018-01-12	772.675000
2018-01-15	714.993889

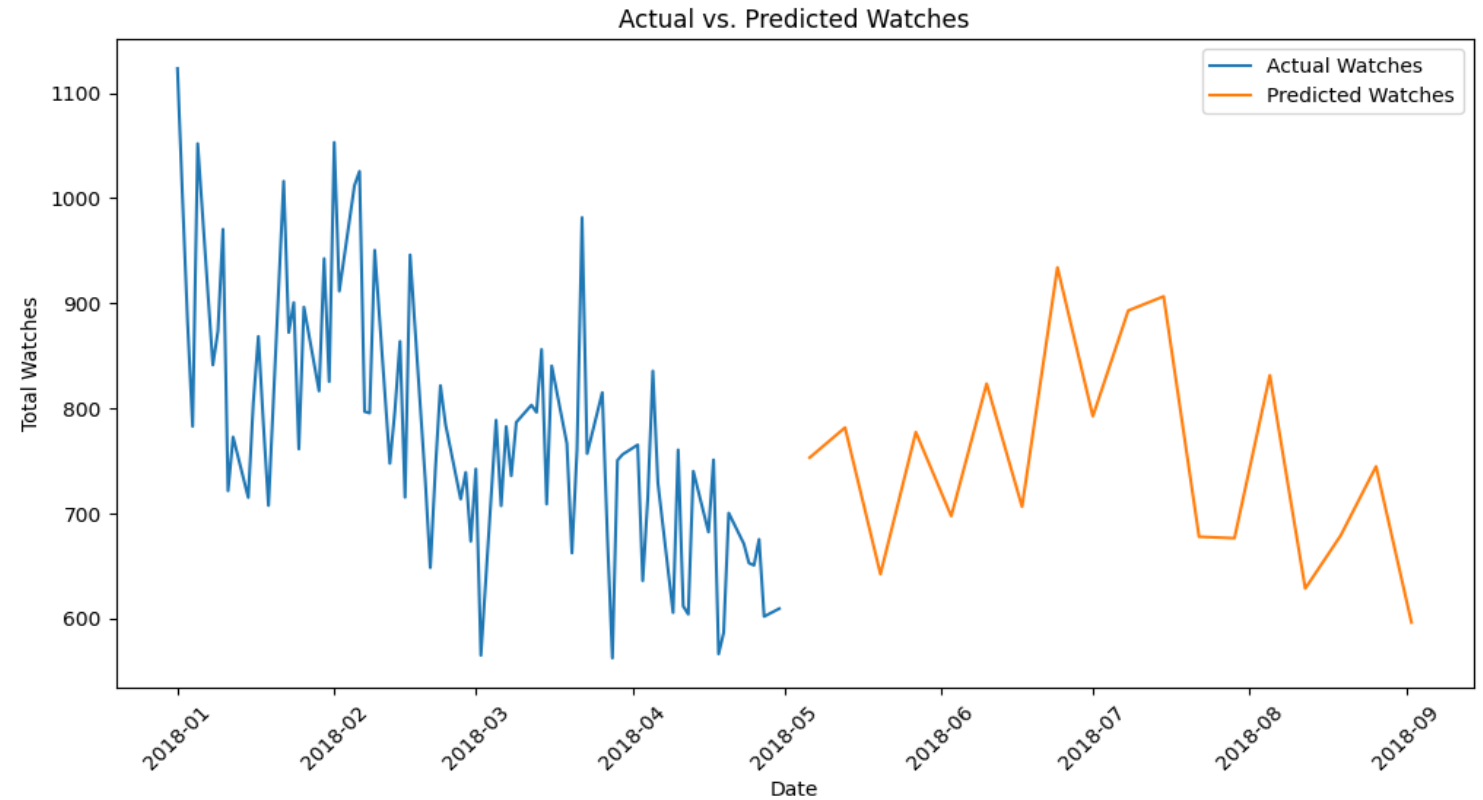
- The dataset shows the total watching hours for customers per day over the first four months of 2018, as depicted in this image.



- Our task is to predict the total watch time for the next two months.
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- First, we need to understand the underlying patterns and behaviors within the data. Typically, a time series is decomposed into three components: trend, seasonality, and noise (or residual/remainder), as depicted in this image.



After that, we trained a model to predict the watch time for the next months. The results of the actual watch time and the predictions are shown in this image.



Third Task

- We have a dataset that includes details about each customer, the movies and/or TV shows they have watched, and the genres of those movies and shows.
- Our task is to build a recommendation engine for our platform to suggest movies to users that they might be interested in.

	user_id_mapped	program_name	rating	date_	program_genre
0	26138	100 treetts	1	2017-05-27	Drama
1	7946	Moana	1	2017-05-21	Animation
2	7418	The Mermaid Princess	1	2017-08-10	Animation
3	19307	The Mermaid Princess	2	2017-07-26	Animation
4	15860	Churchill	2	2017-07-07	Biography
5	20775	Beavis And Butt-Head Do America	1	2017-08-19	Animation
6	11374	The Mermaid Princess	4	2017-06-26	Animation
7	6358	Coco	1	2018-03-29	Animation
8	11660	Kidnap	2	2018-01-27	Action
9	5155	The Accountant	1	2017-03-30	Action
10	30889	The Birth of a Nation	2	2017-04-10	Biography
11	1035	The Amazing pider-Man	2	2017-08-30	Action
12	12598	Baywatch	1	2018-02-02	Comedy
13	7805	Alien: Covenant	2	2017-09-24	Horror
14	33846	Going in tyle	2	2017-09-20	Comedy
15	12333	Kingsman: The Golden Circle	3	2018-01-20	Action
16	25343	Billy Elliot the Musical Live	1	2018-02-10	Comedy

- First, we will build a recommender system to highlight programs that users might be interested in based on the `weighted_rating()` formula.
- Then, we will calculate the number of users rating each program and the average of all the votes for that program.

		vote_count	vote_average
program_name	program_genre		
#FollowFriday	Thriller	510	1.784314
10 Days in a Madhouse	Thriller	553	1.707052
100 treet	Drama	1851	1.836305
101 Dalmatians	Family	118	1.466102
102 Dalmatians	Action	101	1.564356
	Comedy	71	1.549296
11.6	Drama	379	2.018470
12 Monkeys Arms Of Mine	Drama	51	3.176471
12 Monkeys Atari	Drama	82	3.317073
12 Monkeys Blood Washed Away	Drama	28	3.642857

- To define the score of the program, we use the `weighted_rating()` formula.
- $$WR = (v / (v + m)) * R + (m / (v + m)) * C$$
- Where:
- **WR**: Weighted rating
- **R**: Average rating for the show
- **v**: Number of ratings for the show
- **m**: Minimum number of ratings required to be considered (a threshold)
- **C**: Average rating across all shows

- Now, we can rearrange the dataset by sorting the scores of the programs from highest to lowest.

	program_name	program_genre	vote_count	vote_average	score
0	PAW Patrol Pups ave A chool Day	Animation	809	3.610630	3.422205
1	PAW Patrol Pups And The Beanstalk	Animation	810	3.600000	3.414271
2	PAW Patrol Pups ave The Easter Egg Hunt	Animation	818	3.591687	3.409303
3	PAW Patrol Pups Go All Monkey / Pups ave A Hoot	Animation	791	3.595449	3.407434
4	PAW Patrol Pups And The Lighthouse Boogie	Animation	775	3.596129	3.405017
5	PAW Patrol Pups ave A Hoedown / Pups ave Alex	Animation	834	3.575540	3.399586
6	PAW Patrol Pups ave A Pool Day / Circus Pup F...	Animation	821	3.576127	3.397903
7	PAW Patrol Pups ave The Camping Trip	Animation	787	3.583227	3.397449
8	PAW Patrol Pups ave A uper Pup	Animation	782	3.565217	3.382924
9	PAW Patrol Pups Pit Crew / Pups Fire Fight	Animation	1013	3.518263	3.378610

Finally, we can filter the previous dataset to show the top 5 recommendations for people who watched the movie “Moana,” based on similarities such as program genre.

index		program_name	program_genre	vote_count	vote_average	score
0	0	PAW Patrol Pups ave A chool Day	Animation	809	3.610630	3.422205
1	1	PAW Patrol Pups And The Beanstalk	Animation	810	3.600000	3.414271
2	2	PAW Patrol Pups ave The Easter Egg Hunt	Animation	818	3.591687	3.409303
3	3	PAW Patrol Pups Go All Monkey / Pups ave A Hoot	Animation	791	3.595449	3.407434
4	4	PAW Patrol Pups And The Lighthouse Boogie	Animation	775	3.596129	3.405017

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- *Data Scientist : Sana'a Salem Saleem Aloufi*